

Infection

AFIAS Anti-HBs

INTENDED USE

AFIAS Anti-HBs is a fluorescence Immunoassay (FIA) for the qualitative determination of antibody to Hepatitis B surface antigen (anti-HBs) in human whole blood/serum/plasma. It is useful as an aid in determination of susceptibility to HBV infection or following Hepatitis B virus vaccination.

For *in vitro* diagnostic use only.

INTRODUCTION

Viral hepatitis is a serious global health problem affecting over two billion people worldwide and approximately one million people die each year due to cirrhosis of the liver and hepatocellular carcinoma (HCC), which are commonly associated with chronic hepatitis. The majority of hepatitis viral infections are caused by three distinct virus types: Hepatitis A (HAV), Hepatitis B (HBV) and hepatitis C (HCV).^{1,2)} The risk of developing chronic infection by HBV varies inversely with age and is highest for infants infected at birth compared to older children and adults. Up to 90% of infants infected with HBV will develop chronic infection leading to cirrhosis of the liver or HCC compared to 6-10% of adults who acquire HBV infection.³⁾ Determination of antibodies directed against HBV surface antigen (anti-HBs) is used to evaluate a person's immune status to HBV infection or to aid in the laboratory diagnosis of HBV infection when used in conjunction with other laboratory methods. The test is performed to assess the need for vaccination (if anti-HBs is absent or below levels considered protective), following completion of vaccination against HBV in high risk groups (healthcare workers, Chronic renal failure patients, HIV infected persons), or to monitor recovery from acute HBV infection. The presence of anti-HBs following acute infection generally indicates recovery and immunity from reinfection.

PRINCIPLE

The test uses a sandwich immunodetection method; dried detector antigen in cartridge binds to anti-HBs in sample, forming antigen-antibody complexes, and migrates onto nitrocellulose matrix to be captured by the other immobilized-antigen on test strip.

The more anti-HBs in sample forms the more antigen-antibody complex and leads to stronger intensity of fluorescence signal on detector antigen, which is processed by the instrument for AFIAS test to show anti-HBs concentration in sample.

COMPONENTS

AFIAS Anti-HBs consists of 'Cartridge', 'Pipette tip', 'ID chip' and 'Instruction for use'.

- Each cartridge packaged in an aluminum pouch has three components, a detector part, cartridge part and diluent part.
- Cartridge part contains a test strip, the membrane which has recombinant HBsAg at the test lines, while chicken IgY at the control line.
- Detector part contains dried detection buffer and granulated ball. It contains recombinant HBsAg fluorescence conjugate, anti-chicken IgY fluorescence conjugate, bovine serum albumin (BSA) as a stabilizer and sodium azide in phosphate buffered saline (PBS) as a preservative.
- Diluent part contains bovine serum albumin (BSA) as a stabilizer and sodium azide in phosphate buffered saline (PBS) as a preservative.

WARNINGS AND PRECAUTIONS

- For *in vitro* diagnostic use only.
- Carefully follow the instructions and procedures described in this 'Instruction for use'.
- Use only fresh samples and avoid direct sunlight.
- Lot numbers of all the test components (cartridge and ID chip) must match each other.
- Do not interchange the test components between different lots or use the test components after the expiration date, either of which might yield misleading of test result(s).
- The cartridge should remain sealed in its aluminum pouch until use. Do not use the cartridge if that is damaged or already opened.
- For shipping, samples must be packed in accordance with the regulations. Sample with severe hemolytic and hyperlipidemia cannot be used and should be recollected.
- Just before use, allow the cartridge and sample to be at room temperature for approximately 30 minutes.
- AFIAS Anti-HBs** as well as the instrument for AFIAS test should be used away from vibration and/or magnetic field. During normal usage, it can be noted that the instrument for AFIAS test may produce minor vibration.
- Used pipette tips, and cartridges should be handled carefully and disposed by an appropriate method in accordance with relevant local regulations.
- An exposure to larger quantities of sodium azide may cause certain health issues like convulsions, low blood pressure and heart rate, loss of consciousness, lung injury and respiratory failure.
- AFIAS Anti-HBs** will provide accurate and reliable results subject to the following conditions.
 - AFIAS Anti-HBs** should be used only in conjunction with the instrument for AFIAS test.
 - Any anticoagulants other than EDTA, heparin, sodium citrate should be avoided.

LIMITATIONS OF THE TEST SYSTEM

- The test may yield false positive result(s) due to the cross-reactions and/or non-specific adhesion of certain sample components to the capture/detector antigens.
- The test may yield false negative result. The non-responsiveness of the antibodies to the antigens is most common where the epitope is masked by some unknown components, so as not to be detected or captured by the antigens. The instability or degradation of the antibody with time and/or temperature may cause the false negative as it makes antibody unrecognizable by the antigens.
- Other factors may interfere with the test and cause erroneous results, such as technical/procedural errors, degradation of the test components/reagents or presence of interfering substances in the test samples.
- Any clinical diagnosis based on the test result must be supported by a comprehensive judgment of the concerned physician including clinical symptoms and other relevant test results.

STORAGE AND STABILITY

- The cartridge is stable for 20 months (while sealed in an aluminum pouch) if stored at 2-8 °C.
- Return an unused cartridge to the spare cartridge zipperbag containing the desiccant pack. Reseal along entire edge of zip-seal. May be stored for up to 1 month at 2-8°C.

MATERIALS SUPPLIED

REF SMFP-16

Components of **AFIAS Anti-HBs**

- Cartridge Box Contains

- Cartridge	24
- Pipette Tip (Zipperbag)	24
- Spare Cartridge Zipperbag	1
- ID Chip	1
- Instruction For Use	1

MATERIALS REQUIRED BUT SUPPLIED ON DEMAND

Following items can be purchased separately from **AFIAS Anti-HBs**. Please contact our sales division for more information.

- AFIAS-1** **REF** FPRR019
- AFIAS-6** **REF** FPRR020
- Boditech Anti-HBs Control** **REF** CFPO-144

SAMPLE COLLECTION AND PROCESSING

The sample type for **AFIAS Anti-HBs** is human whole blood/serum/plasma.

- It is recommended to test the sample within 24 hours after collection.
- The serum or plasma should be separated from the clot by centrifugation within 3 hours after the collection of whole blood.
- Samples may be stored for up to a week at 2-8 °C prior to being tested. If testing will be delayed more than a week, serum or plasma sample should be frozen at -20 °C.
- Serum or plasma sample stored frozen at -20 °C for 2 months showed no performance difference.
- However, the whole blood sample should not be kept in a freezer in any case.
- Once the sample was frozen, it should be thawed one time and only for test, because repeated freezing and thawing can result in the changed test values.

 **Samples containing precipitates must be clarified by centrifugation.**

TEST SETUP

- Check the components of the **AFIAS Anti-HBs** as described below. : Cartridge, pipette tip, ID chip and instruction for use
- Keep the sealed cartridge (if stored in refrigerator) at room temperature for at least 30 minutes just prior to the test. Place the cartridge on a clean, dust-free and flat surface.
- Turn on the instrument for AFIAS test.
- Empty the tip box.
- Insert the ID chip into the "ID chip port".
- Please refer to the instrument for AFIAS test 'Operation Manual' for complete information and operating instructions.

TEST PROCEDURE

- Select 'General Mode' in the instrument for AFIAS test
- Take 100 µl of sample with a pipette and dispense it into the sample well on the cartridge.
- Insert the cartridge into the cartridge holder.
- Insert a tip in to the tip hole of the cartridge.
- Input 'Patient ID' and select 'Sample type' into the instrument for AFIAS test.
- Tap the 'START' icon on the screen.
- Tap the 'Confirm start' icon on the screen after reconfirmation of 'Patient ID' and 'Sample type'
- The test results will be displayed on the screen after 15 minutes.

※ Note: Refer to the instrument for AFIAS test Operation Manual to select sample type.

INTERPRETATION OF TEST RESULT

- the instrument for AFIAS test calculates the test result automatically and displays "Positive"/ "Negative"/ "Indeterminate".
- Ancillary value is served in form of 'mIU/mL'.

Value (mIU/mL)	Result	Note
≤ 5	Negative for anti-HBs	No need to additional test
5 < Titer < 15	Indeterminate	Need to retest
≥ 15	Positive for anti-HBs	Need to confirmation test

QUALITY CONTROL

- Quality control tests are a part of the good testing practice to confirm the expected results and validity of the assay and should be performed at regular intervals.
- The control tests should be performed immediately after opening a new test lot to ensure the test performance is not altered.
- Quality control tests should also be performed whenever there is any question concerning the validity of the test results.
- Control materials are not provided with **AFIAS Anti-HBs**. For more information regarding obtaining the control materials, contact **Boditech Med Inc.'s Sales Division for assistance**.
(Please refer to the instruction for use of control material.)

PERFORMANCE CHARACTERISTICS

Analytical sensitivity

- The detection limit of **AFIAS Anti-HBs** is 5.784 mIU/mL.

KFDA STD		Result of AFIAS Anti-HBs	
Concentration (mIU/mL)	Value (mIU/mL)	Result	CV (%)
48.8	58.36	Positive	1.5
24.4	25.64	Positive	4.0
12.2	11.9	Indeterminate	5.3
9.2	8.92	Indeterminate	5.6
6.9	7.12	Indeterminate	7.0
5.1	5.32	Indeterminate	8.8
3.9	3.27	Negative	26.3
2.9	2.957	Negative	59.1

* CV; Coefficient of Variation (%)

- Cut-off: 15 mIU/mL

Analytical specificity

Cross-reactivity

There was no cross- reactions with clinical category such as Cytomegalovirus (CMV), Epstein-Barr virus (EBV), Hepatitis A virus (HAV), Hepatitis C virus (HCV), Herpes simplex virus (HSV), Rubella virus, Varicella-zoster virus (VZV), *Treponema pallidum*, Anti Nuclear Antibody (ANA), Rheumatoid factor (RF), and Prenatal tests with the **AFIAS Anti-HBs** test measurement.

Interference

There was no interferences were observed when materials were added in the test samples with the **AFIAS Anti-HBs**.

Materials	Concentration
Heparin	100,000 U/L
EDTA	4 μM
Sodium citrate	25 mg/ml
Hemoglobin	2 mg/mL
BSA	60 mg/mL
Bilirubin	400 μM
Triglycerides	1.5 mg/mL

Cholesterol	20 mM
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Precision

[Between Lot] One person tested three different lots of **AFIAS Anti-HBs**; ten times at each concentration of the control standard.

[Between persons] Three persons tested one lot of **AFIAS Anti-HBs**; five times at each concentration of the control standard.

[Between day] One person tested one lot of **AFIAS Anti-HBs** during three days; five times at each concentration of control standard.

[Between site] One person tested **AFIAS Anti-HBs** at three different site; five times at each concentration of the control standard.

Cal.	Between LOT		Between person		Between day		Between site	
	Positive /sample number	Positive	Positive /sample number	Positive	Positive /sample number	Positive	Positive /sample number	Positive
Negative	0/10	0 %	0/5	0 %	0/5	0 %	0/5	0 %
High	10/10	100 %	5/5	100 %	5/5	100 %	5/5	100 %
Mid	10/10	100 %	5/5	100 %	5/5	100 %	5/5	100 %
Low	10/10	100 %	5/5	100 %	5/5	100 %	5/5	100 %

Comparability

AFIAS Anti-HBs	Reference Anti-HBs assay		
	Positive	Negative	Total
	Positive	118	12
	Negative	4	672
	Total	122	684

- Percent positive agreement: 96.7 %
- Percent Negative agreement: 98.2 %
- Overall percent agreement: 98.0 %

REFERENCES

- The Global Burden of Liver Disease: The Major Impact of China. *Hepatology*. 2014, 60: 2099-2108.
- Viral hepatitis in resource-limited countries and access to antiviral therapies: current and future challenges. *Future Virol*. 2013, 8: 371-380.
- Mahoney FJ, et al. Update on diagnosis, management and prevention of hepatitis B virus infection, 1999 Clin Microbiol Rev, 12:351-366.

Note: Please refer to the table below to identify various symbols.

	Sufficient for <n> tests
	Read instruction for use
	Use by Date
	Batch code
	Catalog number
	Caution
	Manufacturer
	In vitro diagnostic medical device
	Temperature limit
	Do not reuse

For technical assistance; please contact:

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